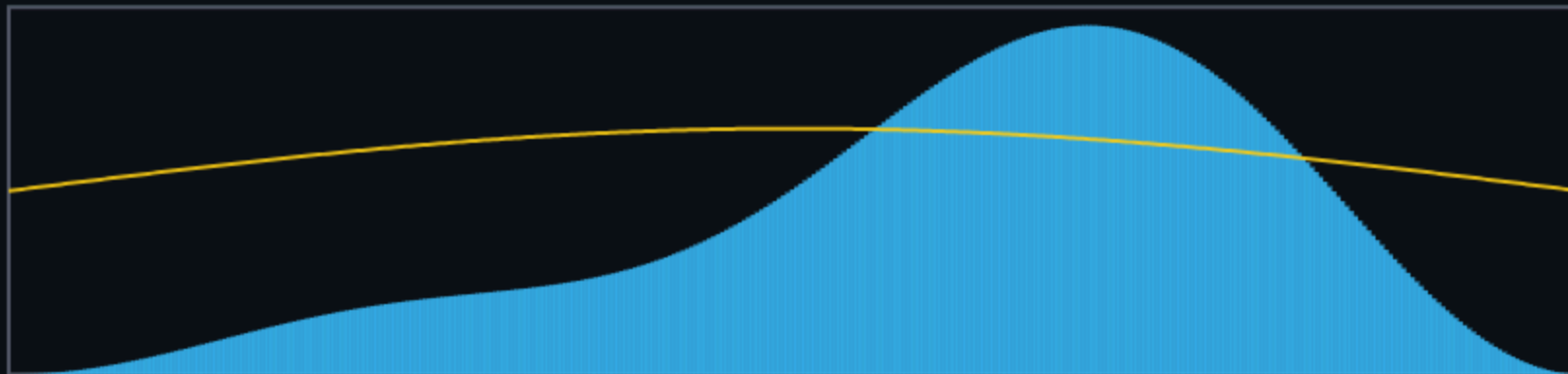


Particle in a 1D Box

Superposition of $n=1$ and $n=2$: time evolution of $|\Psi|^2$ and $\langle E \rangle$ (model units).



SUPERPOSITION N=1,2

Amplitude c_1 (mode 1)

0.75

Amplitude c_2 (mode 2)

0.55

Time speed

1x

Reset time

Stationary states $\psi_n \propto \sin(n\pi x/L)$ with $E_n \propto n^2$.
A superposition $\Psi = c_1\psi_1 + c_2\psi_2$ beats because $|\Psi|^2$ has cross terms oscillating at frequency $(E_2 - E_1)/\hbar$ (here in model units).

MEASURED VALUES

$\langle E \rangle$ (relative units)

2.049

About this model

The Particle in a 1D Box is a foundational quantum mechanical model that illustrates the wave-like nature of matter. This simulator visualizes a crucial non-stationary state: the time-dependent superposition of the ground ($n=1$) and first excited ($n=2$) energy eigenstates. The total wavefunction is $\Psi(x,t) = c_1 \psi_1(x) e^{-iE_1 t/\hbar} + c_2 \psi_2(x) e^{-iE_2 t/\hbar}$, where $\psi_n(x) = \sqrt{2/L} \sin(n\pi x/L)$ are the spatial eigenfunctions for an infinite potential well of length L , and $E_n = n^2 \pi^2 \hbar^2 / (2mL^2)$ are the corresponding quantized energies. The simulator plots the time evolution of the probability density, $|\Psi(x,t)|^2$, which shows a sloshing motion as the relative phase between the two components changes. It also tracks the expectation

value of energy, $\langle E \rangle = |c_1|^2 E_1 + |c_2|^2 E_2$, which remains constant in time for this closed system, demonstrating energy conservation. Key simplifications include the infinite well walls (perfect confinement, no tunneling), a single spatial dimension, and the omission of relativistic effects. By interacting with this model, students learn to interpret superposition states, observe how interference between eigenstates leads to a dynamic probability density, and verify that while the wavefunction evolves, the energy expectation value for a superposition of stationary states is time-independent—a direct consequence of the Schrödinger equation.

Who it's for: Undergraduate students in introductory quantum mechanics or physical chemistry courses learning about wavefunctions, superposition, and time evolution.

KEY TERMS

- Quantum Superposition
- Wavefunction
- Probability Density
- Stationary State
- Expectation Value
- Schrödinger Equation
- Energy Eigenstate
- Time Evolution

How it works

Infinite square well: discrete levels and spatial nodes. Interference between two energy eigenstates makes the probability density slosh back and forth — the quantum analog of beats.

Frequently asked questions

Why does the probability density $|\Psi|^2$ slosh back and forth, but the average energy $\langle E \rangle$ doesn't change?

The sloshing arises from the interference between the $n=1$ and $n=2$ states, which have different time-dependent phase factors. This changes the shape of Ψ , and thus $|\Psi|^2$, over time. However, $\langle E \rangle$ is a weighted average of the fixed energies E_1 and E_2 . Since the weights ($|c_1|^2$ and $|c_2|^2$) are constant, $\langle E \rangle$ is also constant, illustrating energy conservation for a system not interacting with its environment.

Is a particle really in two energy states at once? What does superposition mean?

Yes, in quantum mechanics, a particle can exist in a superposition of states. This doesn't mean it's sometimes in $n=1$ and sometimes in $n=2$. Rather, it is in a new, combined state described by the sum Ψ . When you measure the energy, you will collapse the system to either E_1 or E_2 with probabilities $|c_1|^2$ and $|c_2|^2$. Before measurement, the particle has no definite energy from this set; it possesses the property of the superposition state itself.

What are the main limitations of the infinite 1D box model?

The infinite walls are an idealization; real potential wells have finite height, allowing for quantum tunneling. The model is also one-dimensional and ignores other degrees of freedom (like spin) and interactions between particles. Despite these simplifications, it correctly captures core quantum concepts—quantization, wave-particle duality, and superposition—making it an essential pedagogical tool.

Does this have any connection to real-world systems?

Absolutely. While idealized, this model approximates the behavior of electrons confined in nanostructures like quantum dots or in certain organic molecules with conjugated pi-electron systems (e.g., butadiene). In these systems, electrons are effectively confined to a region of space, and their allowed energies are quantized, influencing optical and electrical properties.

Related simulators

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Quantum Well Eigenstates (Box & HO)

▶ Launch Simulator

Stationary ψ_n and $|\psi_n|^2$ for infinite square well or harmonic oscillator; optional phase animation ($\hbar = 1$).

NEW UNIVERSITY / RESEARCH

Hydrogen $|\psi|^2$ (xz slice)

▶ Launch Simulator

Coulomb hydrogen $Z=1$: $|R_{nl}|^2 \times |Y_{lm}|^2$ colormap in the xz plane for textbook n,l states.

NEW SCHOOL

Orbital Shapes (Schematic)

▶ Launch Simulator

2D $|\psi|^2$ colormap for s-, p-, and d-like angular patterns (pedagogical, not HF).

NEW UNIVERSITY / RESEARCH

2D Box: Eigenstates & Degeneracy

▶ Launch Simulator

Particle in a 2-D rectangular infinite well: $\psi_{\{n_x,n_y\}} \propto \sin(n_x\pi x/L_x)\sin(n_y\pi y/L_y)$, $E \propto (n_x/L_x)^2 + \dots$

NEW SCHOOL

Maxwell–Boltzmann vs E_a

▶ Launch Simulator

Translational energy density $f(E)$; shaded fraction above activation energy; compare Arrhenius $\exp(-E_a/RT)$.

NEW SCHOOL

Reaction Rate

▶ Launch Simulator

Arrhenius $k(T)$, order $0/1/2$, $[A](t)$ curves and particle tank.